

Residual Minimization And Scientific ML Reliability — cycles 7-9

2026-05-14

Contents

Residual Minimization And Scientific ML Reliability — cycles 7-9	1
Abstract	1
Introduction	1
Approach	2
Findings	3
Discussion	5
Open Questions	5
References	6
Appendix: Implementation Details	6

Residual Minimization And Scientific ML Reliability — cycles 7-9

Abstract

Cycles 7-9 moved the residual-minimization reliability run from validated branch work into final synthesis and closure. Cycle 7 produced and validated the M-12 broad synthesis package, which integrated the catalogue, application map, branch notes, toy simulations, certificates, stability baselines, and remaining limitations. Cycle 8 converted that package into a reader-facing final report and validated the final delivery artifact. Cycle 9 confirmed that no new research branch was needed and pivoted the process from further research/audit cycling to session closure.

The final supported thesis is narrow: a residual objective certifies physical behavior only when it is coercive, or otherwise matched, for the physical behavior being claimed. The package does not claim that physics-informed neural networks, neural operators, collocation methods, or residual minimization fail in general. It records explicit objective-function failures, matched stability baselines, and the certificates that distinguish them.

Introduction

The project studies residual minimization in scientific machine learning. A residual loss measures how well a candidate function, learned vector field, parameter, or surrogate satisfies an equation or constraint. The central question is whether a small residual loss certifies the intended physical behavior: convergence in the correct norm, admissibility, stability, conservation, observability, parameter identifiability, or deployment behavior.

Earlier cycles built the validated ingredients: a 22-entry residual-case catalogue, an application risk map, branch reports for weak topology, admissibility/invariants, and ODE reliability, and a toy simulation suite. Cycles 7-9 did not introduce a new mathematical mechanism. They consolidated those ingredients into the final closure package and checked that the package met the breadth and reproducibility goals in the directive.

The references used by the package connect the examples to physics-informed neural networks [1], least-squares and residual methods [2], numerical PDE stability and approximation theory [3], scattered-data and sampling ideas [4], and conservation-law entropy selection [5].

Approach

The report uses the nine provided cycle sessions as primary sources:

Cycle	Role	Session ID	Main content
7	Researcher	3129ecc7-e873-4b3e-bb95-667a2c85d367	Defined M-12 as the synthesis task and set sufficiency criteria.
7	Worker	0fc2ccc8-b5f0-46b7-a846-5a44bf7a8733	Built residual-certificates/broad_synthesis_package.md.
7	Auditor	e6621bb0-1fa1-410e-80fd-288900903590	Validated M-12.
8	Researcher	1eafe17e-a909-415e-9f30-e4c005513bd6	Defined final delivery packaging.
8	Worker	afb1d91b-c93a-4bf2-8f04-4ed9a8d03643	Built reports/residual_minimization_reliability_final_report.md.
8	Auditor	c691389f-da2b-448d-9521-a0a2dd0a4f22	Validated final delivery.
9	Researcher	ed428912-b3b6-4875-b8a1-f7530b9fa68a	Recommended closure and no further build/run cycle.
9	Worker	690c2936-0421-4a84-b3ec-c17d1afe76dd	Confirmed no new artifact was built because the package was complete.
9	Auditor	d8242068-00ea-4b2e-87f0-f391300162dc	Confirmed checks passed and issued a PIVOT decision toward session closure.

The sequence is chronological. Cycle 7 asked whether the validated work met the directive's breadth criteria. Cycle 8 asked whether a human reader could use and reproduce the package without reconstructing the ledger. Cycle 9 asked whether any further research cycle was justified.

Findings

Cycle 7: Broad Synthesis

Cycle 7 closed M-12 by building and validating `residual-certificates/broad_synthesis_package.md`. The researcher session (3129ecc7-e873-4b3e-bb95-667a2c85d367) framed M-12 as an integration task, not a discovery task. The key question was whether the validated artifacts collectively met the directive's breadth goals without overclaiming unresolved M-2 and M-4 labels.

The worker session (0fc2ccc8-b5f0-46b7-a846-5a44bf7a8733) created the synthesis package and made narrow consistency updates to:

- `residual-certificates/toy_simulation_results.md`
- `residual-certificates/ode_reliability_branch.md`
- `residual-certificates/application_risk_map.md`
- `promise_ledger.jsonl`

The synthesis recorded the following breadth counts:

Criterion	Recorded count
Attempted mechanisms	22
Explicit objective or family entries	14+
Theorem-quality failures or rigorous obstructions	10+
Toy simulations or variants	10

The synthesis organized the project around mechanism families: finite sampling and collocation noncoercivity, trace leakage, weak topology mismatch, quadrature aliasing, admissibility and entropy gaps, hidden modes, deployment-region mismatch, inverse identifiability, eigenmode ambiguity, and stability baselines under matched norms.

The worker ran the full focused toy suite:

```
35 passed in 61.80s
```

The worker also ran `promise_check` and `org_check`, both with exit code 0. The remaining warnings were historical or intentionally preserved: M-2 and M-4 lacked standalone ledger events, old orphan/session/report artifacts remained, and root-layout warnings applied to existing launch/config scripts.

The auditor session (e6621bb0-1fa1-410e-80fd-288900903590) validated M-12. The audit checked the synthesis package against the sufficiency criteria, cross-checked source coverage against the catalogue, application map, toy results, ODE branch, and ledger, and spot-checked 23 referenced CSV/PNG artifacts with no missing files. The audit reran the full toy tests:

```
35 passed in 57.56s
```

The auditor appended an M-12 validation event to `promise_ledger.jsonl` and concluded that the package closed the broad synthesis requirement without claiming generic failure of residual minimization.

Cycle 8: Final Delivery Packaging

Cycle 8 converted the validated synthesis package into a compact reader-facing report. The researcher session (1eafe17e-a909-415e-9f30-e4c005513bd6) defined the task as final delivery packaging. The explicit goal was to make the validated package usable by a human reader while preserving the limitations around M-2, M-4, CAT-18, and CAT-20.

The worker session (afb1d91b-c93a-4bf2-8f04-4ed9a8d03643) created:

- reports/residual_minimization_reliability_final_report.md

The report included the final thesis, validated artifact inventory, mechanism/certificate evidence table, toy evidence summary, reproducibility commands, and explicit limitations. It indexed the validated package rather than reopening scientific branches.

The worker ran the full focused toy suite:

```
35 passed in 60.42s
```

The worker also ran `promise_check` and `org_check`, both with exit code 0. The final report preserved the key limitations:

- no generic claim that PINNs or residual methods fail;
- no production-system failure claim;
- M-2 remains pending / partly absorbed;
- M-4 remains partly absorbed by Burgers/admissibility analysis but not a full shock-selection simulation campaign;
- CAT-18 and CAT-20 remain deferred.

The auditor session (c691389f-da2b-448d-9521-a0a2dd0a4f22) validated the final delivery report. The audit checked the final report, ledger, limitation language, failure/baseline labels, and reproducibility commands. It reran the focused toy suite:

```
35 passed in 52.88s
```

The audit also confirmed:

- `promise_check`: exit 0, warnings only;
- `org_check`: exit 0, warnings only;
- final appended ledger event parses as valid JSON.

The auditor appended validation event 6e5eaf51-90a2-4580-9715-c2ec1d2679c2 for `_run/final-delivery-ready` at 2026-05-14T17:31:00Z.

Cycle 9: Closure And Pivot

Cycle 9 confirmed that the package was complete and that another research branch was not warranted. The researcher session (ed428912-b3b6-4875-b8a1-f7530b9fa68a) stated that no new research questions remained inside the current directive. Future work, if reopened, should target only the explicitly labeled loose ends:

- standalone M-2;
- standalone M-4;
- CAT-18;
- CAT-20.

The worker session (690c2936-0421-4a84-b3ec-c17d1afe76dd) built no new artifact. It restated the validated final artifacts:

- reports/residual_minimization_reliability_final_report.md
- residual-certificates/broad_synthesis_package.md
- residual-certificates/residual_case_catalogue.md
- residual-certificates/application_risk_map.md
- branch notes, scripts, tests, CSVs, and PNGs

The auditor session (d8242068-00ea-4b2e-87f0-f391300162dc) confirmed that the final report existed, promise_check and org_check passed with warnings only, and the ledger tail contained _run/final-delivery-ready validated by the auditor. The audit decision was PIVOT, not VALIDATED, because cycle 9's worker output contained no new artifact, build, test, or scientific result. The pivot was procedural: stop research cycling and move to session closure.

Discussion

Cycles 7-9 completed the transition from research package to final deliverable. The project's final claim is a certificate statement: residual losses certify physical behavior only when the loss controls the target behavior. When the loss has a nullspace, uses the wrong topology, omits a boundary or admissibility condition, projects away hidden state, lacks excitation, underintegrates a residual, or covers only the wrong deployment region, it can become misleading. When the objective is matched to the target behavior, classical estimates and constraints can restore reliability.

The completed package separates failures from baselines. For example, weak negative-norm objectives can miss strong physical error, but matched elliptic residuals in the correct dual norm are stability baselines. Equality residuals can miss entropy or positivity, but maximum and comparison principles prevent certain continuous-residual failures under exact data. ODE residuals can miss hidden modes, deployment instability, or unidentifiable parameters, but scalar stiff continuous residuals with fixed initial data are controlled by variation of constants.

By the end of cycle 9, the validated milestones included M-1, M-3, M-5, M-6, M-7, M-8, M-9, M-10, M-11, M-12, and final delivery packaging. M-2 and M-4 remained pending or partly absorbed rather than silently closed.

Open Questions

The run is ready for human use, but the following items remain explicitly outside the validated closure:

- M-2: a standalone continuous norm-mismatch theorem remains pending. Its substance is partly represented by weak-topology and trace-leakage results.
- M-4: a standalone conservation-law selection campaign remains partly absorbed. The Burgers entropy example is present as a precise admissibility proof sketch, but not as a full shock-selection simulation campaign.
- CAT-18: inverse PDE or source recovery through sensor-nullspace mechanisms remains deferred.
- CAT-20: long-horizon surrogate drift remains deferred and would need a clean setup separate from the validated one-step and Lyapunov examples.

The cycle 9 guidance was not to reopen any of these unless the user explicitly asks for a new branch.

References

- [1] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, 378, 686-707, 2019. <https://doi.org/10.1016/j.jcp.2018.10.045>
- [2] P. B. Bochev and M. D. Gunzburger, "Least-Squares Finite Element Methods," *Applied Mathematical Sciences* 166, Springer, 2009. <https://link.springer.com/book/10.1007/b13382>
- [3] A. Quarteroni and A. Valli, "Numerical Approximation of Partial Differential Equations," *Springer Series in Computational Mathematics* 23, Springer, 1994. <https://link.springer.com/book/10.1007/978-3-540-85268-1>
- [4] H. Wendland, "Scattered Data Approximation," *Cambridge Monographs on Applied and Computational Mathematics* 17, Cambridge University Press, 2005. <https://www.cambridge.org/core/books/scattered-data-approximation/966D4683385F92FB8D1535F8D0A9585E>
- [5] R. J. LeVeque, "Finite Volume Methods for Hyperbolic Problems," *Cambridge Texts in Applied Mathematics*, Cambridge University Press, 2002. <https://www.cambridge.org/core/books/finite-volume-methods-for-hyperbolic-problems/CB7B0A27A6D37AE3B906D4AE7C60A708>

Appendix: Implementation Details

Code Organization

The final package is organized around four main reader-facing artifacts:

- `residual-certificates/broad_synthesis_package.md`: validated M-12 synthesis with breadth counts, mechanism taxonomy, closure table, toy-suite summary, and limitations.
- `reports/residual_minimization_reliability_final_report.md`: validated final report for human consumption and reproduction.
- `residual-certificates/residual_case_catalogue.md`: 22-entry mechanism catalogue.
- `residual-certificates/application_risk_map.md`: application mapping with evidence-strength labels.

The supporting branch notes are:

- `residual-certificates/weak_topology_branch.md`
- `residual-certificates/admissibility_invariant_branch.md`
- `residual-certificates/ode_reliability_branch.md`
- `residual-certificates/collocation_blind_spot_theorem.md`

The toy suite is implemented in `scripts/`, with matching tests in `tests/` and CSV/PNG outputs in `data/`.

Test Results

Cycle 7 worker validation:

```
35 passed in 61.80s
promise_check: exit 0
org_check: exit 0
```

Cycle 7 auditor validation:

```
35 passed in 57.56s
promise_check: exit 0, warnings only
org_check: exit 0, warnings only
```

Cycle 8 worker validation:

```
35 passed in 60.42s
promise_check: exit 0
org_check: exit 0
```

Cycle 8 auditor validation:

```
35 passed in 52.88s
promise_check: exit 0, warnings only
org_check: exit 0, warnings only
```

Cycle 9 auditor check:

```
promise_check: exit 0, warnings only
org_check: exit 0, warnings only
```

The remaining warnings are historical or intentionally preserved: old noncanonical artifact paths, M-2/M-4 lacking standalone ledger events, orphan/session artifacts, missing old manager-assessment artifacts, missing old reports/final/final_report.md, and root-level script/config organization warnings.

File Counts

The current manifest records:

Group	Files	Lines
Scripts/check scripts	12	1341
Tests	11	554
Residual-certificate notes	13	998
Final/report manifest files	3	175
Root reference/ledger files	2	57
Total tracked files	41	3125

MANIFEST.md was updated during this reporting pass to reflect the final-package state.

Session References

Cycle 7:

- Researcher: 3129ecc7-e873-4b3e-bb95-667a2c85d367
- Worker: 0fc2ccc8-b5f0-46b7-a846-5a44bf7a8733
- Auditor: e6621bb0-1fa1-410e-80fd-288900903590

Cycle 8:

- Researcher: 1eafe17e-a909-415e-9f30-e4c005513bd6

- Worker: afb1d91b-c93a-4bf2-8f04-4ed9a8d03643
- Auditor: c691389f-da2b-448d-9521-a0a2dd0a4f22

Cycle 9:

- Researcher: ed428912-b3b6-4875-b8a1-f7530b9fa68a
- Worker: 690c2936-0421-4a84-b3ec-c17d1afe76dd
- Auditor: d8242068-00ea-4b2e-87f0-f391300162dc

Cross-Reference Map

Claim or package element	Source artifact
Final coercivity/certificate thesis	residual-certificates/broad_synthesis_package.md; reports/residual_minimization_reliability_final_report.md
22-mechanism catalogue	residual-certificates/residual_case_catalogue.md
Application mapping	residual-certificates/application_risk_map.md
Toy simulation evidence	residual-certificates/toy_simulation_results.md; scripts/; tests/; data/
Weak topology branch	residual-certificates/weak_topology_branch.md
Admissibility and invariant branch	residual-certificates/admissibility_invariant_branch.md
ODE reliability branch	residual-certificates/ode_reliability_branch.md
Final human-readable package	reports/residual_minimization_reliability_final_report.md
Final validation ledger	promise_ledger.jsonl